Revision | **Covers** 3-1 → 3-4 | **R** Lab Quiz I | **File** 6 / 8

REV

Revision pack: formulas, decisions, traps, R, A+ plan

حزمة المراجعة — القوانين، اختيار المقياس، الأخطاء، أوامر R، وخطة A+

1 · Master formula sheet

NAME	FORMULA	USE / REMEMBER
3-1 · Central tendency		
Mean المتوسط	$\bar{x} = \Sigma X/n \quad \mu = \Sigma X/N$	uses all values; affected by outliers; not for open-ended classes
Grouped mean	$\bar{x} = \Sigma f \cdot X_m / n$	X_m = class midpoint
Median الوسيط	position $(n+1)/2$; even: mean of $n/2$ and $n/2 + 1$	order first; grouped: first class with CF > N/2
Mode المنوال	most frequent value	none / uni / bi / multimodal; grouped: modal class
Midrange منتصف المدى	$MR = (\text{lowest} + \text{highest})/2$	very rough; outlier-sensitive
Weighted mean المتوسط المرجّح	$\bar{x} = \Sigma wX / \Sigma w$	unequal importance (GPA, course grade, combining group means)
3-2 · Variation		
Range المدى	$R = \text{highest} - \text{lowest}$	a single number
Population variance / SD	$\sigma^2 = \Sigma(X - \mu)^2/N \quad \sigma = \sqrt{\sigma^2}$	whole population: divide by N
Sample variance / SD	$s^2 = \Sigma(X - \bar{x})^2/(n - 1) \quad s = \sqrt{s^2}$	sample: divide by n − 1
Shortcut	$s^2 = [n(\Sigma X^2) - (\Sigma X)^2] / [n(n - 1)]$	also $(\Sigma X^2 - n \cdot \bar{x}^2)/(n - 1)$

NAME	FORMULA	USE / REMEMBER
Grouped	$s^2 = [n(\Sigma f \cdot X_m^2) - (\Sigma f \cdot X_m)^2] / [n(n-1)]$	columns D = $f \cdot X_m$, E = $f \cdot X_m^2$
Coefficient of variation معامل الاختلاف	$CVar = (s/\bar{x}) \cdot 100$ $(\sigma/\mu) \cdot 100$	no units; compare across units or means; smaller = more consistent
Range rule of thumb	$s \approx R/4$	unimodal and symmetric only; usually underestimates
Chebyshev نظرية تشيبيشيف	at least $1 - 1/k^2$ within $\bar{x} \pm ks$	any shape; k = (limit − mean)/SD; k = 2: 75%, k = 3: 88.89%
Empirical rule القاعدة التجريبية	≈ 68% · 95% · 99.7% within 1 · 2 · 3 SD	bell-shaped only
Linear transformation	$y = a \cdot x + b$	centre → a·c + b; SD, IQR → ×|a|; variance → ×a^2
3-3 · Position		
z-score الدرجة المعيارية	$z = (X - \bar{x})/s$ $z = (X - \mu)/\sigma$	$X = \bar{x} + z \cdot s$; all z-scores: mean 0, SD 1
Percentile rank رتبة المئين	[(# below X) + 0.5]/n · 100	order the data first
Value at P_p	$c = n \cdot p/100$	decimal: round up and take that value · whole: average of c-th and (c + 1)-th
Quartiles · deciles	$Q_1 = P_{25}$, $Q_2 = P_{50} = MD = D_5$, $Q_3 = P_{75}$; $D_k = P_{10k}$	median method or percentile formula
IQR · outlier fences	$IQR = Q_3 - Q_1$; $Q_1 - 1.5 \cdot IQR$, $Q_3 + 1.5 \cdot IQR$	outside the fences = outlier
3-4 · EDA		
Five-number summary الملخص الخماسي	min, Q_1, MD, Q_3, max	in this order; it draws the boxplot
Shape from a boxplot	median position + whisker lengths	median left and right whisker long → right skew

2 · Which tool? Ten-second decisions

THE QUESTION SAYS…	USE
"all employees", "the entire population", "every …"	μ, σ^2, σ, dividing by **N**
"a sample of", "a random sample", "selected"	$\bar{x}$, s^2, s, dividing by **n − 1**
nominal data (colour, blood type, brand)	**mode** only
outliers, skew, income or salary data, open-ended class	**median** (and IQR)
values of unequal importance; combining group means	**weighted mean**
"which is more variable / consistent?" with different units or means	**CV**, never raw SDs
"what can be said about the number / percentage…", shape unknown	**Chebyshev**: "at least"
"bell-shaped", "normal", "mound-shaped"	**empirical rule**: "approximately"
"compare relative position / who did better relative to their group"	**z-scores**
"what percentile is X?" / "what value is at P_p?"	percentile formula / c = n·p/100
"check for outliers", "unusual value"	fences $Q_1 - 1.5 \cdot IQR$, $Q_3 + 1.5 \cdot IQR$
"add 5 to every score" / "convert units"	linear transformation rules

3 · Fifteen traps that cost marks

1. Finding the median or quartiles **without ordering** the data first.
2. Dividing a **sample** variance by n, or a **population** variance by n − 1.
3. Treating ΣX^2 as $(\Sigma X)^2$.
4. Rounding deviations mid-calculation (the 2.335 vs 2.333 lesson).
5. Writing "mode = 0" instead of "no mode".
6. Computing a mean for a table with an **open-ended class**.
7. Comparing raw SDs when the means or units differ; use the CV.
9. Writing "exactly 75%" for Chebyshev; it is **at least**. Round counts **up** for "at least".
10. Using the empirical rule when the shape is not stated as bell-shaped.
11. Percentile rank: counting values ≤ X instead of strictly **below** X, or forgetting + 0.5.
12. c = n·p/100 with a decimal c: rounding to the nearest number instead of **up**.
13. Including the median in both halves when n is odd (course method).
14. Thinking a longer boxplot section holds more data; every section is 25%.

8. Forgetting to take the √ when the question gives the **variance**.
15. "Adding a constant changes the SD": it does not. Multiplying changes the variance by c^2, not c.

4 · R for Lab Quiz I (outline CLO2-D, CLO3-D)

TASK	R	WATCH OUT
mean, median	`mean(x)` `median(x)`	use `na.rm = TRUE` if there are missing values
mode	`names(which.max(table(x)))`	`mode(x)` is **not** the statistical mode; it returns the storage type
weighted mean	`weighted.mean(x, w)`	
range	`diff(range(x))` or `max(x) - min(x)`	`range(x)` returns the min **and** max, not R
variance, SD	`var(x)` `sd(x)`	**sample** formulas (n – 1)
CV	`sd(x)/mean(x)*100`	
z-scores	`(x - mean(x))/sd(x)` or `scale(x)`	
quartiles, IQR, summary	`quantile(x, c(.25,.5,.75))` `IQR(x)` `summary(x)`	interpolated: may differ from the hand method
five-number summary, boxplot	`fivenum(x)` `boxplot(x, horizontal = TRUE)`	Tukey hinges; outliers drawn as points

5 · The A+ plan for Chapter 3

DAY 1 **Lessons 3-1 and 3-2, one solved example per formula.**
Close the file and redo Ex 3-20/3-21 both ways (definition and shortcut) until both give 13.1 without notes. That one example covers half the variance marks in the chapter.

DAY 2 **Lessons 3-3 and 3-4.**
Use Jim's 20 scores as your master data set: percentile rank, P_{50}, P_{74}, quartiles by both methods, IQR, fences, boxplot. One data set, every position skill.

DAY 3 **Interpretation drill (CLO3).**

For every number you computed, write one sentence of meaning: "more consistent", "at least 75%", "positively skewed because...". This is where students who calculate perfectly still lose marks.

DAY 4 **The question bank**

, closed book, then check against the answers file. Textbook page 178, Q 2, 3, 10, 17 and the Moodle sets: these are the outline's assigned exercises.

QUIZ **In the exam:**

underline "sample" or "population" and "bell-shaped" before you write anything. Order the data before any median or percentile. Finish every answer with one interpretation sentence.

خطة A+ بالعربي

(1) احفظ القوانين بالكتابة من الذاكرة ثم التصحيح. (2) خذ **بيانات Jim العشرين** وطبّق عليها كل مهارات الموقع: المئين، الربيعات بالطريقتين، المدى الربيعي، القيم الشاذة، والمخطط الصندوقي. (3) **درّب نفسك على التفسير**: بعد كل رقم اكتب جملة تشرح معناه، فهذا ما يقيسه مخرج التعلم CLO3. (4) حل بنك الأسئلة مغلق الكتاب، ثم صحّح من ملف الإجابات. (5) في الامتحان: ضع خطًا تحت كلمة **sample** أو **population** وكلمة **bell-shaped** قبل البدء، و**رتّب البيانات** قبل أي وسيط أو مئين.

Chapter 3 study set: 8 files. One merged source built from Dr. Amani Al-Rubkhi's Fall 2026 notes (course coordinator) and Dr. Ahmed Al-Hadhrami's Spring 2024 slides, aligned to the SQU STAT 2101 course outline (Fall 2026). Textbook: Bluman, *Elementary Statistics*, 10th ed.

00 · Exam brief: what the outline tests · 01 · Lesson 3-1: Measures of Central Tendency · 02 · Lesson 3-2: Measures of Variation · 03 · Lesson 3-3: Measures of Position · 04 · Lesson 3-4: Exploratory Data Analysis · **05 · Revision pack: formulas, decisions, traps, R** · 06 · Exam-style questions (no answers) · 07 · Full worked answers